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Exam Questions

Section 1, Module 1: Reading and Writing

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Question 1

Microbiologist Radamés Cordero and team investigated surface temperatures of a variety of fungi, including *Coprinellus micaceus* and a species from the genus *Russula*, in a study _____blank a previous observation: Cordero had taken thermal images of various wild mushrooms while hiking and found that the mushrooms all appeared to be colder than the surrounding environment, raising questions about thermoregulation in fungi.

Which choice completes the text with the most logical and precise word or phrase?

A) prompted by

B) excluded by

C) released in

D) acknowledged in

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Question 2

The _____blank Hyewon Yum's book *Puddle* has been very positive. Many reviewers have praised the book, and it won the Asian/Pacific American Award for Literature.

Which choice completes the text with the most logical and precise word or phrase?

A) response to

B) inspiration for

C) neglect of

D) confusion about

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Question 3

The following text is adapted from Jhumpa Lahiri's 2003 novel *The Namesake*. Ashima and her husband, Ashoke, are traveling from the United States to India.

Ashoke puts his hand over hers as the plane begins to gather speed. And then Boston tilts away and they ascend effortlessly over a blackened Atlantic.

©2003 by Jhumpa Lahiri

As used in the text, what does the word "gather" most nearly mean?

A) Demand

B) Gain

C) Doubt

D) Accept

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Question 4

Research on *Champsocephalus gunnari* and other icefishes of the Teleostei class has shown that their hearts are pale due to low levels of myoglobin expression, which was once thought to be related to their polar habitats. Recently, researchers have found that some tropical teleost species (such as *Piaractus brachypomus*) also have pale hearts, raising the possibility that myoglobin deficiency in teleosts could be _____blank habitat.

Which choice completes the text with the most logical and precise word or phrase?

A) mediated by

B) obscured by

C) irrespective of

D) predictive of

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Question 5

Editor Jared Shurin's 2023 anthology *The Big Book of Cyberpunk* contains 108 stories, an enormous number when compared to the 1986 cyberpunk anthology *Mirrorshades*, which has only twelve. Encompassing both early (e.g., Samuel R. Delany's 1968 story "Time Considered as a Helix of Semi-Precious Stones") and more recent (e.g., Lavanya Lakshminarayan's 2020 story "Études") contributions, the broad chronological scope of Shurin's volume may make it seem more comprehensive than *Mirrorshades*, but *Mirrorshades*'s focus is more carefully considered, rendering it the more authoritative anthology of this science fiction subgenre.

Which choice best describes the function of the underlined portion in the text as a whole?

A) It indicates a sensible perception of a cyberpunk anthology that can nonetheless be disputed for a good reason.

B) It supports the assertion that Delany and Lakshminarayan should be considered essential figures in the cyberpunk movement.

C) It helps explain why two compilers of anthologies have different ideas regarding which short stories should be labeled as "cyberpunk."

D) It suggests that, because they are relatively up to date, more recently published collections of cyberpunk stories are likely to be more inclusive than earlier anthologies.

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Question 6

Yingguang Frank Chan and colleagues have explored how convergent evolution—a phenomenon that occurs when the same trait evolves independently in two reproductively separate lineages—can result from a genetic mechanism shared by both lineages. Meanwhile, Patricia J. Wittkopp and colleagues have investigated how convergence occurs through different genetic mechanisms, but the relative prevalence of convergence through shared and different genetic processes is still poorly understood. This motivated biologists Delbert A. Green II and Cassandra G. Extavour to evaluate both types of convergence in a single study for their 2012 paper.

Which choice best states the function of the underlined portion in the text as a whole?

A) It gives a basic description of a phenomenon that is central to the discussion that follows.

B) It provides examples of how a phenomenon was studied by scientists in the field before Green and Extavour's study.

C) It clarifies a concept that the author implies was unclear in the studies mentioned in the text.

D) It introduces a method of scientific analysis that is discussed in greater detail later in the text.

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Question 7

In 2013 Bridie J.M. Allan and colleagues published a study concluding that ocean acidification has a strong effect on the behavior of *Pomacentrus amboinensis*, a species of fish. However, Allan and colleagues' study relied on a mean sample size of only about 18 fish. In a 2022 review of various scientists' conclusions about the impacts of ocean acidification on fish behavior, Josefin Sundin and colleagues caution that relying on such a relatively small sample size can increase the potential for biased analysis. Such analysis, in turn, can contribute to reports of exaggerated effects.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It emphasizes the threat of ocean acidification to *Pomacentrus amboinensis* that is mentioned in the previous sentence.

B) It introduces a detail about Allan and colleagues' study that is addressed in the sentences that follow.

C) It states the conclusion reached by Allan and colleagues that is elaborated on in the sentences that follow.

D) It offers an alternative explanation for the behavior of *Pomacentrus amboinensis* that is described in the previous sentence.

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Question 8

Did you know that the USS *Merrimac* is just one of around three million known shipwrecks spread across the world's oceans? Rachel Mugge and team wanted to find out how wooden shipwrecks affect the distribution of microbes on the seafloor. The researchers studied two wooden shipwrecks in the Gulf of Mexico. They placed pieces of pine and oak between 0 and 200 meters away from each shipwreck to collect samples of microbes. They found that pine and oak pieces placed about 125 meters from the shipwrecks had the most species of microbes on them.

According to the text, at what approximate distance from the studied shipwrecks did the researchers observe the most species of microbes on the pine and oak pieces?

A) 30 meters

B) 200 meters

C) 0 meters

D) 125 meters

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Question 9

In 1951, Ethel Payne, known as the First Lady of the Black Press, began writing for the esteemed Sengstacke newspapers, which included the *Chicago Defender*. Payne's accomplishment is just one example of the rich history of Black journalism in the United States. That history is preserved by the National Association of Black Journalists, which was founded in Washington, DC, in 1975 to support Black media professionals and honor people like Payne.

According to the text, where was the National Association of Black Journalists founded?

A) Detroit

B) San Diego

C) Washington, DC

D) Philadelphia

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Question 10

Baleen whales eat up to 30 percent of their total body mass in krill (tiny shrimplike creatures) per day. So in one day, a sei whale weighing 25,000 kg could eat 7,500 kg, while a northern right whale weighing 80,000 kg could ingest a whopping 24,000 kg. Over the last century, baleen whale populations have declined, and contrary to some scientists' expectations, so have krill populations. Matthew S. Savoca and colleagues resolve this apparent discrepancy by pointing out that baleen whales cycle iron in the ocean, helping support phytoplankton populations, which, in turn, sustain krill populations.

Based on the text, what is one likely effect of the change in baleen whale populations over the last century?

A) The quantity of krill needed to support a single northern right whale has increased over the last century.

B) Populations of phytoplankton that krill feed on have declined over the last century.

C) The iron richness of the krill that baleen whales eat has declined over the last century.

D) The maximum body sizes of sei and northern right whales have increased over the last century.

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Question 11

Because humans occupy the International Space Station (ISS), it also harbors common dust-borne fungi and bacteria found on Earth. In a study of dust collected from the ISS, the growth of fungi like *Penicillium chrysogenum* and bacteria like *Staphylococcus pettenkoferi* was acceptably low under the station's standard 25%–75% relative humidity (RH) conditions but unacceptably high at RH of 85% and above, which could feasibly happen in station areas where crew members exercise.

Which statement about microbial growth aboard the ISS is best supported by the text?

A) The growth rates of both *P. chrysogenum* and *S. pettenkoferi* increased gradually when RH increased incrementally from 25% to 75%.

B) Neither *P. chrysogenum* nor *S. pettenkoferi* is likely to grow when RH is kept below 25%.

C) Under normal operating conditions, there should be no microbial growth in dust on board the ISS.

D) Undesirable microbial growth in dust likely occurs in some locations on board the ISS.

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Question 12

Average Hours Worked per Person per Year in 1950 and 2017

Country	1950	2017	Change in hours	Percent change in hours
Brazil	2,042	1,709	−333	−16%
Germany	2,427	1,354	−1,074	−44%
Colombia	2,323	1,998	−325	−14%
Portugal	2,399	1,863	−536	−22%

Calculations may be inexact due to rounding.

A student in an economics course is examining the decline since 1950 in average hours worked per person per year in various nations due to both increased productivity and the adoption of policies that limit working hours. The first task in this investigation is to determine how the decline in Portugal compares to that in other countries. The student finds that _____blank

Which choice most effectively uses data from the table to complete the statement?

A) though the percent decrease in hours worked in Portugal was greater than that in Germany and Colombia, it was less than that in Brazil.

B) though the percent decrease in hours worked in Portugal was less than that in Germany, it was greater than that in Brazil and Colombia.

C) while the number of hours worked rose in Portugal from 1950 to 2017, it declined in Germany, Brazil, and Colombia.

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D) the decline in number of hours worked was greater in Portugal than it was in Germany, Brazil, or Colombia.

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Question 13

The Age of Innocence is a 1920 novel by Edith Wharton set in New York City in the 1870s. In the novel, Newland Archer attends an opera. Newland compares his intellect favorably to that of other men of New York City society who are in the audience: _____blank

Which quotation from *The Age of Innocence* best illustrates the claim?

A) "It surprised [Newland] that life should be going on in the old way when his own reactions to it had so completely changed."

B) "Singly [the men around Newland] betrayed their inferiority; but grouped together they represented 'New York,' and the habit of masculine solidarity made him accept their doctrine on all the issues called moral."

C) "But, in the first place, New York was a metropolis, and perfectly aware that in metropolises it was 'not the thing' to arrive early at the opera."

D) "Newland Archer felt himself distinctly the superior of these chosen specimens of old New York gentility; he had probably read more, thought more, and even seen a good deal more of the world, than any other man of the number."

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Question 14

Paleontologist Amane Tajika and colleagues analyzed the shells of two marine mollusks called nautilids that were harvested from New Caledonia. The researchers found that samples from shell sections that formed before hatching, including sample F07, formed in 20–22°C water, whereas samples from sections that formed after hatching, including M13, formed in 13–15°C water. These findings are consistent with the temperatures Fijian nautilids encounter during their corresponding stages of development. Because water temperature varies with depth, a biology student hypothesized that the New Caledonian nautilids lay their eggs at approximately the same depth as Fijian nautilids do.

Which finding, if true, would most directly weaken the student's hypothesis?

A) Fijian nautilids generally live at lower depths than New Caledonian nautilids do, though Fijian nautilids migrate to the typical depth of New Caledonian nautilids to lay their eggs.

B) Once a nautilid hatches, it is a very mobile animal, so the temperature values reported by Tajika and colleagues for the adult nautilids are averages over a potentially large range of depths.

C) The water temperature at a given depth tends to be lower in waters near New Caledonia than it is at that depth in waters near Fiji.

D) There are few locations near Fiji with water temperatures of 13–15°C that are suitable for nautilid eggs.

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Question 15

In 1952, the Swedish Academy presented François Mauriac of France with an _____blank that few authors would turn down: the Nobel Prize in Literature.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) award;

B) award,

C) award:

D) award

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Question 16

Which choice most logically completes the text?

A) identify conservation strategies that may not be exclusively aimed at protecting *M. glochidiata* but that benefit the species nonetheless.

B) lead to a better understanding of how the distribution of *Mammillaria* species throughout Mexico has affected their survival.

C) be unable to evaluate whether *M. glochidiata* would be capable of surviving in habitats outside of Hidalgo.

D) allow conservationists to better consider how climatic changes affecting Hidalgo may reduce the number of species competing with *M. glochidiata*.

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Question 17

China fir plants' adaptive response to drought conditions—initiated after insufficient water uptake detected by the plants' roots activates hormone-based signaling mechanisms—_____blank responsible for changes to the plants' morphological characteristics.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) is

B) were

C) are

D) being

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Question 18

When first looking at Ojibwe artist Dawn Nichols Walden's woven baskets, viewers are likely to notice the baskets' unconventional shapes. Upon closer inspection, Walden's careful construction _____blank apparent.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) become

B) becomes

C) are becoming

D) have become

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Question 19

The Fra Mauro map is a mappa mundi from the fifteenth century. Currently held at the Museo Correr in Venice, Italy, it is one of over 1,000 such world maps from the European Middle Ages still remaining, with the even older Hereford map from the fourteenth century (now held at Hereford Cathedral in Hereford, England) _____blank another one.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) being

B) was

C) is

D) has been

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Question 20

When documenting butterfly diversity at Rupa Lake in Nepal, researcher Bandana Subedi and her colleagues spotted a very rare butterfly of the Nymphalidae family and, after careful examination, knew exactly what species it belonged _____blank *Doleschallia bisaltide*.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) to:

B) to;

C) to

D) to and

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Question 21

The relationship between grassland arthropod populations and agricultural fertilizers containing macronutrients—such as nitrogen, phosphorus, and potassium—was the focus of a 2000 study in the _____blank by the researcher Mark E. Ritchie, the study examined a variety of arthropod specimens, which had been collected with sweep nets.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) United States led

B) United States, led

C) United States. Led

D) United States and led

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Question 22

The World Cup of men's soccer, one of the biggest sporting events on the planet, brought 32 national teams from six continents to the host country, Russia, in 2018. _____blank only 16 teams, all from Europe and the Americas, competed in the 1958 World Cup in Sweden.

Which choice completes the text with the most logical transition?

A) For instance,

B) In contrast,

C) Therefore,

D) In other words,

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Question 23

Until 1992, most Olympic Games included demonstration sports, which did not award medals. _____blank a demonstration of hurling, a ball-and-stick sport that originated in Ireland, was featured at the 1904 Summer Olympics in St. Louis.

Which choice completes the text with the most logical transition?

A) Meanwhile,

B) For example,

C) On the other hand,

D) Someday,

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Question 24

Many places that gained their independence from Spain in the nineteenth century still bear the legacy of the Spanish Empire in their language, including Spanish-speaking Costa Rica, which left the empire in 1821. _____blank which ceased to be part of the Spanish Empire in 1714; Spanish is rarely spoken in Luxembourg in the twenty-first century.

Which choice completes the text with the most logical transition?

A) This contrast is evident in Luxembourg,

B) Another example is Luxembourg,

C) The reverse is true of Luxembourg,

D) The results can be heard in Luxembourg,

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Question 25

While researching a topic, a student has taken the following notes:

- Elizabeth Catlett (1915–2012) was a celebrated African American artist.
- She is best known for creating sculptures and prints that explore the Black experience.
- Catlett's sculpture *Students Aspire* is made of bronze.
- She created this sculpture in 1977.

The student wants to specify the year that Elizabeth Catlett created the sculpture *Students Aspire*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Celebrated artist Elizabeth Catlett is best known for creating sculptures and prints that explore the Black experience.

B) Elizabeth Catlett, a celebrated artist, was born in 1915.

C) Elizabeth Catlett was a celebrated artist who created prints as well as sculptures.

D) Elizabeth Catlett created the bronze sculpture *Students Aspire* in 1977.

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Question 26

While researching a topic, a student has taken the following notes:

- Doyle is the nickname of a *Triceratops* fossil specimen housed at the American Museum of Natural History.
- The American Museum of Natural History is located in New York, New York.
- Doyle lived in the Late Cretaceous period.
- Fafnir is the nickname of a *Triceratops* fossil specimen housed at the Science Museum of Minnesota.
- The Science Museum of Minnesota is located in Saint Paul, Minnesota.
- Fafnir lived in the Late Cretaceous period.

The student wants to contrast the locations of the two specimens. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The Science Museum of Minnesota in Saint Paul, Minnesota, houses Fafnir, a *Triceratops* fossil specimen.

B) Doyle is housed at the American Museum of Natural History in New York, New York, while Fafnir is housed at the Science Museum of Minnesota in Saint Paul, Minnesota.

C) The *Triceratops* fossil specimen Doyle is not the only such specimen currently housed in a museum or institute.

D) Like Doyle, Fafnir is a *Triceratops* fossil specimen that lived in the Late Cretaceous period.

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Question 27

While researching a topic, a student has taken the following notes:

- NASA mission STS-122 took place from February 7, 2008, to February 20, 2008.
- Astronauts Rex Walheim and Stanley Love were crew members of STS-122.
- February 11, 2008: Walheim and Love performed an extravehicular activity (EVA) to install equipment on the space shuttle *Columbus*.
- An EVA (also known as a spacewalk) is an activity done by astronauts outside the spacecraft.
- The duration of their EVA was 7 hours and 58 minutes.
- Theirs is counted among the longest completed spacewalks to date.

The student wants to emphasize the duration of Walheim and Love's EVA. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) During their time as crew members of STS-122, Walheim and Love completed an EVA.

B) The EVA completed by STS-122 crew members Walheim and Love took place from February 7, 2008, to February 20, 2008.

C) Walheim and Love completed an EVA, also known as a spacewalk, in which they left the spacecraft to install equipment on the space shuttle *Columbus*.

D) Lasting 7 hours and 58 minutes, Walheim and Love's EVA is one of the longest spacewalks completed to date.

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Question 1

Wireless headphones and other small electronic devices tend to use batteries that can't be easily taken out and swapped for new ones. Environmental policy researcher Jean-Pierre Schweitzer warns that when these internal batteries stop working, the devices are usually thrown away, becoming harmful _____blank.

Which choice completes the text with the most logical and precise word or phrase?

A) advice

B) positions

C) waste

D) rewards

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Question 2

The following text is from Rudolfo Anaya's 1972 novel *Bless Me, Ultima*. Two of the narrator's adult brothers have decided to move away from home. Another brother has decided not to accompany them.

In the morning León and Eugene were gone, but Andrew remained. They had talked long into the night, and finally he had given up the idea of going with them. I think he did not like to follow their ways, and he wanted to please my mother.

©1972 by Rudolfo Anaya

As used in the text, what does the word "follow" most nearly mean?

A) Mention

B) Imitate

C) Retrieve

D) Justify

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Question 3

In addition to the giant stock exchanges in places like London, Mumbai, and Tokyo, there are many smaller exchanges such as the Somali Stock Exchange in Mogadishu, Somalia. Though small, by being dedicated to Somalia, it can better address factors that help local businesses _____blank, and that focus on the success of local businesses benefits investors and the local economy.

Which choice completes the text with the most logical and precise word or phrase?

A) overspend

B) thrive

C) disperse

D) ebb

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Question 4

Daniel J. Kliebenstein and colleagues tested whether soil microorganisms from ecologically different settings—ponderosa pine forest, irrigated pasture, and so on—affect plants' flavor chemistry. Although they found associations between plants' glucosinolates (compounds that produce spicy and bitter flavors) and plants' growth conditions, variations in glucosinolates could not be _____blank soil microbes specifically.

Which choice completes the text with the most logical and precise word or phrase?

A) attributed to

B) subverted by

C) excluded from

D) forestalled by

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Question 5

Text 1

Columbia University and the online class provider edX are two of the many institutions offering training programs in entrepreneurship. But what results do such programs produce? In a study of college students in the United States, Korea, China, and Fiji, researcher Sang Lee and colleagues addressed this question and convincingly demonstrated that participants showed high knowledge of entrepreneurship as a result of receiving entrepreneurial training.

Text 2

While studies of entrepreneurial training typically report positive results, a close look at these studies reveals widespread methodological shortcomings. For instance, a 2005 study led by Sang Lee found benefits of entrepreneurial education for college students in four countries, but the study only collected data on students after the training, not before, making it unclear whether the training actually had an effect.

Based on the texts, the author of Text 1 and the author of Text 2 would most likely disagree about the answer to which question?

A) Did Lee and colleagues report that participants who received the entrepreneurial training showed high knowledge of entrepreneurship?

B) Are there convincing experimental results showing that different types of entrepreneurial training programs are associated with different outcomes?

C) Is the cost of conducting entrepreneurial training programs typically offset by the programs' benefits?

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D) Did Lee and colleagues' study provide persuasive evidence that its results are attributable to the entrepreneurial training participants received?

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Question 6

The following text is adapted from Henrik Ibsen's 1899 play *When We Dead Awaken* (translated by William Archer in 1907). Ulfheim is introducing himself to Professor Rubek, a sculptor.

ULFHEIM: I believe I have the honour of addressing no less a person than the great Sculptor Rubek.

PROFESSOR RUBEK: (*Nods.*) I remember meeting you once or twice—the autumn when I was last at home.

ULFHEIM: That's many years ago, now. And then you weren't so illustrious as I hear you've since become.

Which choice best describes the function of the underlined portion in the text?

A) It suggests that Ulfheim prefers autumn to spring, summer, or winter.

B) It reveals that Professor Rubek and Ulfheim first met many years ago.

C) It indicates that many other sculptors are more famous than Professor Rubek.

D) It emphasizes that Professor Rubek has a distinguished reputation.

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Question 7

Cedar Hill in Washington, DC, is the house where Frederick Douglass lived for nearly twenty years and wrote one of his autobiographies. Now the house is a museum dedicated to Douglass's life and writings. Visitors travel to this museum to feel more connected to an author they admire. To encourage that sense of connection, the museum's curators have set up exhibits including a re-creation of his personal study. Such exhibits help visitors understand day-to-day life in the house where Douglass felt inspired.

Which choice best states the main purpose of the text?

A) To summarize Douglass's biography

B) To describe a museum focused on Douglass's life and works

C) To compare Cedar Hill with other museums

D) To explain why readers enjoy Douglass's works

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Question 8

When used to examine paintings, nonvisible light, such as ultraviolet, can penetrate the painting's visible surface and give art historians important insights into an artist's process. For example, imaging the underlayers of Artemisia Gentileschi's *Saint Catherine of Alexandria* (1618–20) revealed that Gentileschi originally painted her own face on the figure. Imaging can also provide insights into a painter's choice of materials, showing, for example, whether a painter used titanium oxide pigment in a given work.

According to the text, why are the images beneath the surface of a painting valuable to art historians?

A) They can show how long a painting took to complete.

B) They can indicate whether a painting is damaged.

C) They can be used to prove which artist painted a given work.

D) They can reveal aspects of an artist's decision-making.

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Question 9

One challenge faced by researchers studying global urbanization is that different countries have different definitions of what constitutes an urban settlement. Many countries define urban settlements based on the number of people living in them. However, countries do not all use the same number; for example, Iceland uses a minimum population of 200, while Mali uses 30,000. Other countries use a combination of population and other factors, like the availability of planned public spaces. This variation makes it difficult for researchers to compare aspects of urbanization in different countries.

Which choice best states the main idea of the text?

A) Urbanization is easily compared between countries when they have similar total populations.

B) Even though countries use different methods to define urban settlements, researchers consider any area having a population of at least 30,000 to be urban.

C) It is difficult for researchers to compare aspects of urbanization globally because of the various ways different countries define urban settlements.

D) Researchers focus on factors other than population when conducting studies on urbanization in Iceland.

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Question 10

Brown Bears in Katmai National Park, Alaska

Bear identification number	Sex	Age (years)	Approximate weight (pounds)
176	male	10	575
192	female	17	300
149	male	12	750
123	female	11	350

Scientists collected information about brown bears in Katmai National Park in Alaska. This information included each bear's sex, age, and approximate weight. Among the bears whose information is shown in the table, the youngest one was the bear with identification number _____blank

Which choice most effectively uses data from the table to complete the statement?

A) 176.

B) 123.

C) 192.

D) 149.

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Question 11

Year That Foreign Investors Were First Allowed to Purchase Shares

Country	Year
Malaysia	1987
Indonesia	1989
Jamaica	1991

In the 1980s and 1990s, many countries began allowing foreign investors to purchase shares of companies based in those countries.

According to the table, in what year did Malaysia make the change mentioned in the text?

A) 1987

B) 1983

C) 1998

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D) 1981

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Question 12

Psychologists Gregory Bryant, Dorsa Amir, and colleagues investigated cross-cultural perceptions of spontaneous (real) laughter and volitional (fake or forced) laughter. Study participants from 21 societies, including those in Australia and Iran, listened to randomized recordings of 18 spontaneous laughs taken from natural conversations between pairs of women and 18 volitional laughs produced separately by 18 different women in response to an instruction to laugh. Analysis of the participants' evaluations of the laughs prompted the team to conclude that the ability to distinguish between spontaneous and volitional laughter appears to be universal across cultures.

Which potential finding from another study, if true, would most directly strengthen the team's conclusion?

A) There is some evidence of correlation between the average degree of English fluency in a society and judgments of spontaneous and volitional laughter by listeners in that society.

B) Listeners from both Australia and Iran tend to be able to determine how well two people know each other based on recordings of their conversation.

C) Spontaneous laughter typically has a recognizably higher pitch than volitional laughter, regardless of what society the speakers in a conversation are from.

D) Acoustic features of spontaneous laughter, such as pitch and intensity, have measurable variations both within and across societies.

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Question 13

Some representations of the cataclysmic fate of the comic book planet Krypton illustrate a common trope in disaster stories: people's self-destructive carelessness. Explanations of real-world societal collapse betray a similar pattern. One theory about the downfall of Rapa Nui, for example, argues that residents invited disaster by despoiling the island's Jubaea palm forests, leading to a scarcity of wood for fuel and canoe construction. Archaeological evidence, however, suggests that residents adapted to environmental conditions by relying on grass and other plants for cooking and driftwood for making canoes. Given that other factors—e.g., disease and conflict arising from European contact—played a crucial role in Rapa Nui's eventual demise, it therefore seems unlikely that _____blank

Which choice most logically completes the text?

A) deforestation was merely indirectly responsible for an irreversible collapse of Rapa Nui's subsistence economy.

B) its residents' focus on environmental challenges prevented them from devoting adequate attention to political issues.

C) its residents' lack of environmental awareness was ultimately responsible for Rapa Nui's collapse.

D) its residents would have failed to appreciate the relevance of fictional disaster stories to their own predicament.

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Question 14

Many Indigenous songs serve as repositories of ecological knowledge—e.g., Sakha songs about local ecosystems and O'odham songs about desert plants. In a study by ethnobiologist Dana Lepofsky et al., a song keeper for the Kwakwaka'wakw people provided access to ancient songs referring to the people's creation of terraced intertidal gardens to foster clams for consumption. Cross-references of that information with both archaeological and paleoecological records have revealed significant increases in clam size and productivity concurrent with the most prevalent use of these clam gardens. This finding suggests that researchers should consider _____blank

Which choice most logically completes the text?

A) the possibility that other songs of other Indigenous peoples transmit ecological information that has useful practical applications.

B) the reliability of archaeological and paleoecological evidence that often underpins conclusions drawn from Indigenous songs.

C) whether the Kwakwaka'wakw people originated the method of cultivation detailed in the songs or adapted methods observed among other groups of Indigenous peoples.

D) the strong implication that the beneficial effects are primarily attributable to ecological factors unrelated to the activities of the Kwakwaka'wakw people.

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Question 15

The smoke point of refined avocado oil _____blank around 520 degrees Fahrenheit. When refined avocado oil is heated past this temperature, it will begin to combust and emit smoke.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) is,

B) is

C) is;

D) is—

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Question 16

Introducing a new species into a delicate ecosystem can often be harmful. The zebra dove and other fruit-eating birds have established themselves on the Hawaiian island of Oahu after being brought there in the last 150 years. Many plant species on the island rely on the activity of fruit-eating birds to maintain and increase their populations. In recent studies, zebra doves have been found to spread seeds of native forest plants that are at risk of extinction, such as the mamaki shrub and the alahee tree, suggesting that _____blank

Which choice most logically completes the text?

A) although non-native birds initially damaged the plants in Oahu's ecosystem, they are now proving useful to them.

B) non-native birds have had both positive and negative effects on Oahu's native plants.

C) non-native birds, such as the zebra dove, are unlikely to eat fruits from non-native forest plants on Oahu.

D) rather than harming Oahu's ecosystem, some non-native birds provide an important benefit on Oahu.

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Question 17

Because many of the first cattle ranchers in the US learned the trade from Spanish-speaking Mexican *vaqueros* (cowboys), many ranching terms in the English language come from Spanish. One example is the word "rodeo." It comes from the Spanish word *rodear*. Another example is "cinch." _____blank comes from the Spanish word *cincho*.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) He

B) She

C) It

D) Someone

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Question 18

Located in what is now the United Kingdom, the Taversoe Tuick Chambered Cairn is one of the oldest known structures in the world, _____blank researchers a glimpse into ages long gone.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) provides

B) to provide

C) providing

D) provide

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Question 19

In the Suzuki method, classical pieces like Franz Schubert's "Wiegenlied" are taught to beginning music students as young as three, while students of traditional methods often only _____blank such songs at the age of five.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) to begin learning

B) beginning to learn

C) having begun to learn

D) begin learning

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Question 20

Portobello Road Market has operated for more than a century, making _____blank both a popular shopping destination and an important piece of London's history.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) those

B) these

C) it

D) one

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Question 21

Which choice completes the text so that it conforms to the conventions of Standard English?

A) are

B) was

C) have been

D) is

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Question 22

In the early 1920s, researchers at the US Geological Survey determined the geographic center of Texas using a simple method. First, they attached a string to the middle point of a cardboard cutout of the state and suspended it in the air. _____blank they adjusted the suspension point until the cutout hung perfectly flat. Ultimately, this point was determined to be the center.

Which choice completes the text with the most logical transition?

A) Instead,

B) Then,

C) In other words,

D) First of all,

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Question 23

When the first railroads were built in Australia in the 1800s, a variety of track sizes (gauges) were used across the continent, resulting in what historians call the “railway gauge muddle.” The Victorian Goldfields Railway line in Victoria used broad tracks. _____blank the Hotham Valley Railway line in Western Australia used narrow gauge tracks.

Which choice completes the text with the most logical transition?

A) By contrast,

B) In sum,

C) For instance,

D) Similarly,

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Question 24

While researching a topic, a student has taken the following notes:

- Snow hydrologist Zoe Wellschlager helps maintain automated snow measurement stations throughout the western United States.
- They are called SNOTEL sites.
- There are around 70 sites throughout Alaska.
- There is one at Tokositna Valley.

The student wants to emphasize the number of SNOTEL sites in Alaska. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) One particular SNOTEL site is located at Tokositna Valley in Alaska.

B) Snow hydrologist Zoe Wellschlager helps maintain SNOTEL sites, such as the one at Tokositna Valley, Alaska.

C) SNOTEL sites collect snow measurement data throughout the state of Alaska.

D) In Alaska alone, there are around 70 SNOTEL sites located throughout the state.

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Question 25

While researching a topic, a student has taken the following notes:

- Wind is a renewable energy source.
- A wind farm uses turbines to convert wind into electrical power.
- A wind farm's maximum electrical output is called its nameplate capacity.
- Sweetwater Wind Farm is located in Nolan County, Texas.
- Its nameplate capacity is 586 megawatts.

The student wants to emphasize the location of Sweetwater Wind Farm. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Turbines can be used to generate electricity from a renewable energy source: the wind.

B) Sweetwater Wind Farm has a nameplate capacity of 586 megawatts.

C) Nolan County, Texas, is home to Sweetwater Wind Farm.

D) Nameplate capacity is another name for a wind farm's maximum electrical output.

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Question 26

While researching a topic, a student has taken the following notes:

- The Nga Awa Purua geothermal power plant in New Zealand uses a flash steam system to generate electricity.
- Flash steam technology requires geothermal reservoir temperatures above 180°C.
- The use of flash steam systems worldwide is limited because many geothermal reservoirs aren't hot enough.
- The Beowawe geothermal power plant in Nevada uses a binary cycle system.
- Binary cycle technology can generate energy from lower-temperature (less than 180°C) geothermal reservoirs.
- Binary cycle systems involve higher maintenance costs than flash steam systems.

The student wants to compare the disadvantages of the geothermal systems used at the Nga Awa Purua and Beowawe plants. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Compared with the system used at the Nga Awa Purua plant, the system used at the Beowawe plant has a notable disadvantage: its cost.

B) The system at the Nga Awa Purua plant requires temperatures above 180°C, while the system at the Beowawe plant can operate at lower temperatures.

C) Unlike the Beowawe plant, the Nga Awa Purua plant uses a flash steam system, which requires a less common resource: geothermal reservoirs with temperatures above 180°C.

D) The system used at the Beowawe plant overcomes the temperature limitations of the Nga Awa Purua plant's system but is more costly to maintain.

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Question 27

While researching a topic, a student has taken the following notes:

- *Le Pouce* is an artwork by César Baldaccini.
- It is a forty-foot-tall bronze sculpture of a human thumb.
- It is on display on the grounds of a public art museum in Paris, France.
- *Digital Orca* is a work of art by Douglas Coupland.
- It is a sculpture of a whale made from black and white metal cubes.
- It is on display at a public convention center in Vancouver, Canada.

The student wants to emphasize a similarity between *Le Pouce* and *Digital Orca*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Both *Le Pouce* and *Digital Orca* were created by César Baldaccini.

B) *Le Pouce* and *Digital Orca* are both on display at public locations.

C) *Le Pouce*, but not *Digital Orca*, can be found in Paris, France.

D) *Digital Orca*, a sculpture of a whale made from black and white metal cubes, is on display at a public convention center.

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Question 1

An environmental scientist needs to collect at least 38 samples of red clay soil for analysis. If r represents the number of samples the scientist collects for analysis, which inequality represents this situation?

A) $r \leq 38$

B) $r - 38 \geq 38$

C) $r - 38 \leq 38$

D) $r \geq 38$

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Question 2

$$7j = k + 13m$$

The given equation relates the distinct positive numbers j, k, and m. Which equation correctly expresses j in terms of k and m?

A) $j = 7(k + 13m)$

B) $j = \frac{k}{7} + 13m$

C) $j = k + \frac{13m}{7}$

D) $j = \frac{k+13m}{7}$

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Question 3

The y-intercept of the graph of $y = 5x^2 + 72$ is $(0, y)$. What is the value of y?

Correct Answer:

72

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Question 4

An agricultural scientist needs to collect at least 200 soil samples in a certain area for a study on soil quality. If the scientist has already collected 168 soil samples, what is the minimum number of additional soil samples the scientist needs to collect for the study?

A) 368

B) 12

C) 148

D) 32

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Question 5

The list gives the speeds, in miles per hour, of 7 serves made by a professional tennis player.

$$121, 123, 123, 125, 151, 154, 155$$

What is the mean speed, in miles per hour, of these 7 serves?

Correct Answer:

136

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Question 6

The function f is defined by $f(x) = \frac{x}{5} + 9$. For what value of x does $f(x) = 13$?

A) 20

B) -32

C) 13

D) 22

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Question 7

A triathlon is a multisport race consisting of three different legs. A triathlon participant completed the cycling leg with an average speed of 19.3 miles per hour. What was the average speed, in yards per hour, of the participant during the cycling leg?

$(1 \text{ mile} = 1{,}760 \text{ yards})$

A) $33{,}968$

B) 91.2

C) 57.9

D) $5{,}280$

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Question 8

Line k is defined by $y = 5x + 1$. Line j is parallel to line k in the xy-plane and passes through the point $(0, 2)$. Which equation defines line j?

A) $y = 5x + 2$

B) $y = -2x + 2$

C) $y = -5x + 2$

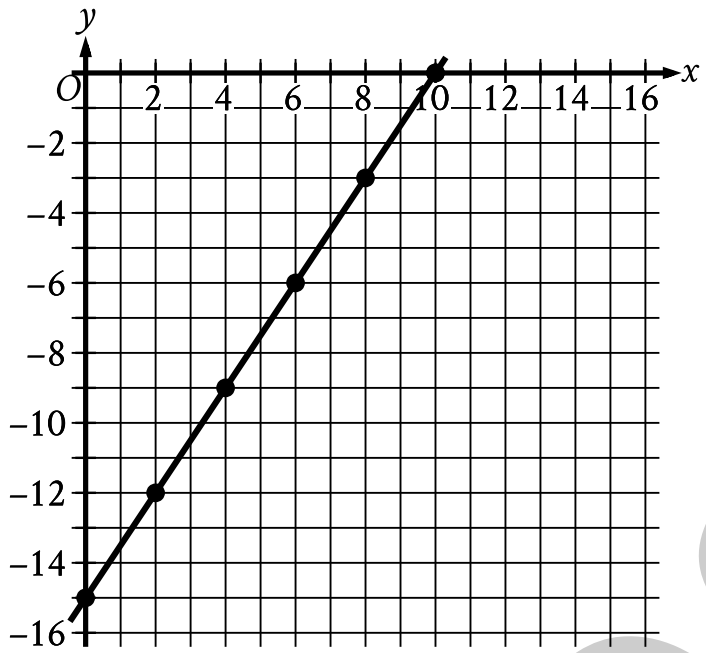
D) $y = 2x + 2$

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Question 9



- The line slants sharply up from left to right.
- The line passes through the following points:
  - (0 comma negative 15)
  - (2 comma negative 12)
  - (4 comma negative 9)
  - (6 comma negative 6)
  - (8 comma negative 3)
  - (10 comma 0)

What is the slope of the line shown?

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Correct Answer:

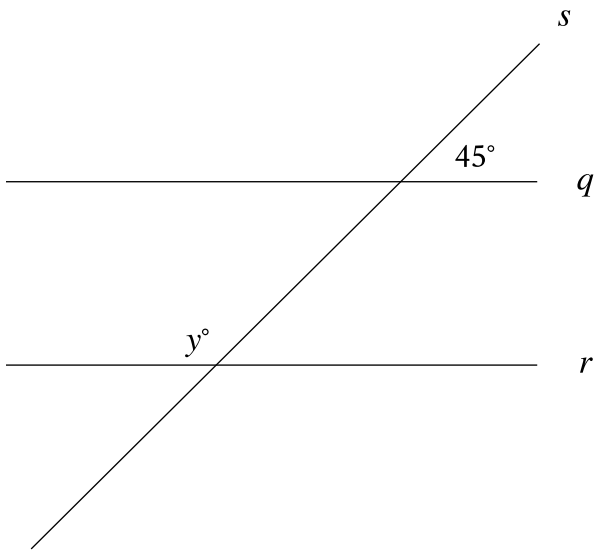
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Question 10



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled s, q, and r.
- Line s intersects both line q and line r.
- At the intersection of line s and line q, 1 angle is labeled clockwise from top left as follows:
  - Top right: 45°
- At the intersection of line s and line r, 1 angle is labeled clockwise from top left as follows:
  - Top left: y°
- A note indicates the figure is not drawn to scale.

In the figure, line q is parallel to line r, and both lines are intersected by line s. If $y = 2x + 9$, what is the value of x?

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A) 27

B) 73

C) 63

D) 18

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Question 11

What is the center of the circle in the *xy*-plane defined by the equation $(x-3)^2+(y+8)^2=9$?

A) $(3,-8)$

B) $(3,8)$

C) $(-3,-8)$

D) $(-3,8)$

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Question 12

$$f(x) = 4{,}000(0.90)^x$$

A conservation scientist implemented a program to reduce a certain invasive insect population in an area. The given function estimates this insect species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of $4{,}000$ in this context?

A) The estimated percent decrease in the insect population for this species and area every 8 years after 2008

B) The estimated initial insect population for this species and area in 2008

C) The estimated percent decrease in the insect population for this species and area each year after 2008

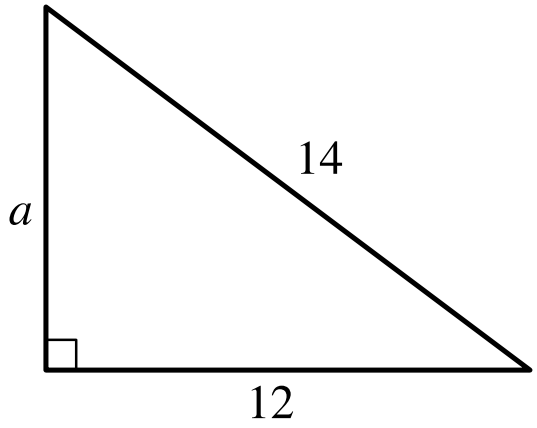
D) The estimated insect population for this species and area 8 years after 2008

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Question 13



Note: . igure not drawn to scale.

- One angle is a right angle.
- The length of the side opposite the right angle is 14.
- The lengths of the other 2 sides are as follows:
  - a
  - 12
- Note: Figure not drawn to scale.

Which equation gives the relationship between the side lengths of the triangle shown?

A) $12^2 + 14^2 = a^2$

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B) $a^2 + 12^2 = 14^2$

C) $a + 12 = 14$

D) $12 + 14 = a$

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Question 14

How many solutions does the equation $14x = 6x + 8x$ have?

A) Exactly one

B) Zero

C) Infinitely many

D) Exactly two

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Question 15

$$\frac{x}{(x-1)(x+1)} = \frac{25}{16x}$$

Which value is a solution to the given equation?

A) $\frac{5}{3}$

B) $\frac{1}{6}$

C) $\frac{5}{18}$

D) 1

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Question 16

A right circular cylinder has a height of 16 inches and a volume of $3{,}136\pi$ cubic inches. What is the radius, in inches, of this cylinder?

Correct Answer:

14

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Question 17

For $x = 0$, y has a value of 200. The value of y increases by 40% for every increase of 1 in the value of x. Which equation represents this situation?

A) $y = 200(x)^{40}$

B) $y = 200(1.40)^{x}$

C) $y = 200(40)^{x}$

D) $y = 200(x)^{1.40}$

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Question 18

$$y = \frac{1}{2}x - 7$$
$$\frac{1}{2}y = 3x + 2$$

The solution to the given system of equations is (x, y). What is the value of $y - x$?

Correct Answer:

–6

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Question 19

$$80x^2 + 80x = -10$$

One solution to the given equation can be written as $x = \frac{-8-k\sqrt{2}}{16}$, where k is a constant. What is the value of k?

A) 4

B) 80

C) 32

D) 128

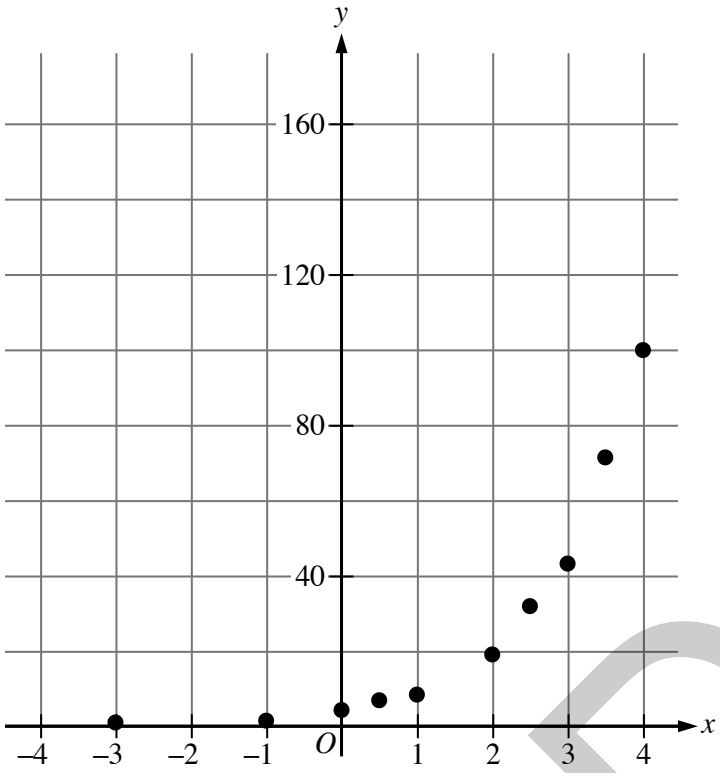
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Question 20

The scatterplot shows the relationship between x and y for the 10 data points in data set A.



- The scatterplot has 10 data points.
- The data points are in an exponential pattern trending up from left to right.
- The data points have the following approximate coordinates:
  - (negative 3 comma 1)
  - (negative 1 comma 1.5)
  - (0 comma 4.3)
  - (0.5 comma 6.9)
  - (1 comma 8.4)

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- (2 comma 19.1)
- (2.5 comma 31.9)
- (3 comma 43.2)
- (3.5 comma 71.3)
- (4 comma 99.8)

Data set B is created by multiplying the y-value of each data point from data set A by 40. Which of the following equations is the most appropriate model for data set B?

A) $y = 5(2.04)^x$

B) $y = 40(2.04)^x + 40$

C) $y = 5(2.04)^x + 40$

D) $y = 199(2.04)^x$

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Question 21

Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D and angle C corresponds to angle F. Angles B and E are right angles. If $\cos A = \frac{14}{\sqrt{421}}$, what is the value of $\frac{1}{\sin F}$?

A) $\frac{15}{\sqrt{421}}$

B) $\frac{14}{\sqrt{421}}$

C) $\frac{\sqrt{421}}{14}$

D) $\frac{\sqrt{421}}{15}$

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Question 22

$$ax - by = 52$$
$$2ax - 8y = 48$$

In the given system of equations, a and b are constants. The graphs of these equations in the *xy*-plane intersect at the point $(x, 4)$. What is the value of b?

A) 3

B) 4

C) -8

D) -3

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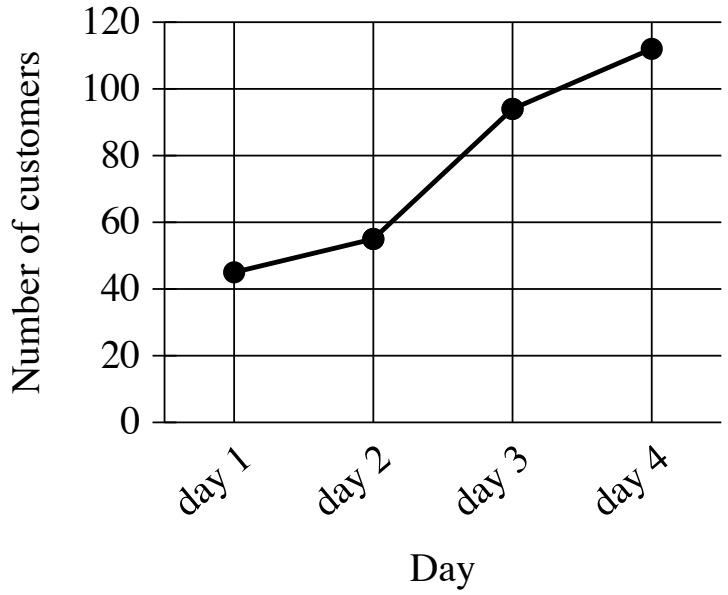
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Question 1



- All values are approximate.
- The line graph:
  - Begins at day 1, 45 customers
  - Rises gradually to day 2, 55 customers
  - Rises sharply to day 3, 94 customers
  - Rises gradually to day 4, 112 customers

The line graph shows the number of customers who visited a restaurant each day during a four-day period in the summer.

According to the line graph, for which day during this four-day period did the restaurant have the smallest number of customers who visited?

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A) Day 3

B) Day 4

C) Day 2

D) Day 1

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Question 2

A designer used a total of 23 yards of blue fabric and yellow fabric for a sewing project. The designer used 17 yards of blue fabric. How many yards of yellow fabric were used?

Enter response...

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Question 3

What is 10% of 450?

A) 35

B) 440

C) 405

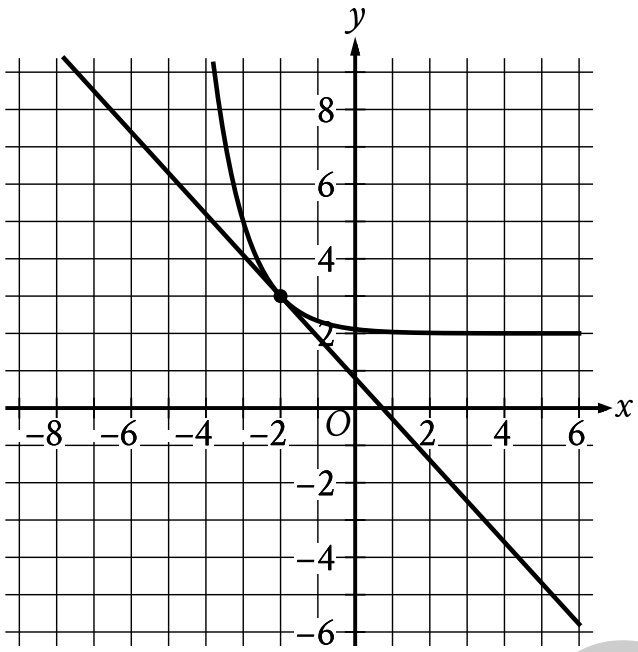
D) 45

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Question 4



- For the line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (negative 2 comma 3)
    - approximately (0 comma 0.8)
- For the curve in the system:
  - Moving from left to right:
    - The curve passes from quadrant 2 to quadrant 1.
    - In quadrant 2:
    - The curve trends down sharply to point (negative 2 comma 3).

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- The curve then trends down gradually to the y axis.
- In quadrant 1, the curve trends down gradually.
  - As x increases, the curve approaches the line y equals 2.
  - The curve passes through the following points:
- (negative 3 comma 5)
- (negative 2 comma 3)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?

A) $(0, 0)$

B) $(0, -2)$

C) $(-2, 3)$

D) $(3, 0)$

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Question 5

$$f(x) = 5x + 800$$

The function shown gives the total cost $f(x)$, in dollars, for making x units of a certain product. What is the total cost, in dollars, for making 100 units of this product?

A) 500

B) 100

C) 800

D) $1{,}300$

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Question 6

Which expression is equivalent to $4x(5xy)$?

A) $20x^2y$

B) $4x^2y$

C) $20x^2y^2$

D) $5x^2y^2$

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Question 7

The length, y, of a beluga whale was 160 centimeters (cm) when it was born and increased an average of 4.7 cm per month for the first 12 months after it was born. Which equation best represents this situation, where x is the number of months after the whale was born?

A) $y = 4.7x + 160$

B) $y = 160x$

C) $y = 4.7x + 4.7$

D) $y = 160x + 160$

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Question 8

$$k^2 - 47 = 97$$

What is the positive solution to the given equation?

A) 144

B) 12

C) 72

D) 50

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Question 9

$$9x + x = 7x + x + 9$$

Which of the following equations has the same solution as the given equation?

A) $16x = -9$

B) $16x = 9$

C) $2x = 9$

D) $2x = -9$

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Question 10

Harper deposited a fixed amount into her bank account each month. The function $f(t) = 400 + 30t$ gives the amount, in dollars, in Harper's bank account after t monthly deposits. What is the best interpretation of 30 in this context?

A) With each monthly deposit, the amount in Harper's bank account increased by $\$30$.

B) Before Harper made any monthly deposits, the amount in her bank account was $\$30$.

C) After 1 monthly deposit, the amount in Harper's bank account was $\$30$.

D) Harper made a total of 30 monthly deposits.

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Question 11

The frequency table summarizes the 105 data values in a data set.

Data value	Frequency
28	13
30	15
31	16
32	17
33	16
34	15
35	13

What is the minimum value of the data set?

Enter response...

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Question 12

What is the *y*-intercept of the graph of $y = x^2 + 27$ in the *xy*-plane?

A) $(0, -27)$

B) $(0, 28)$

C) $(0, 27)$

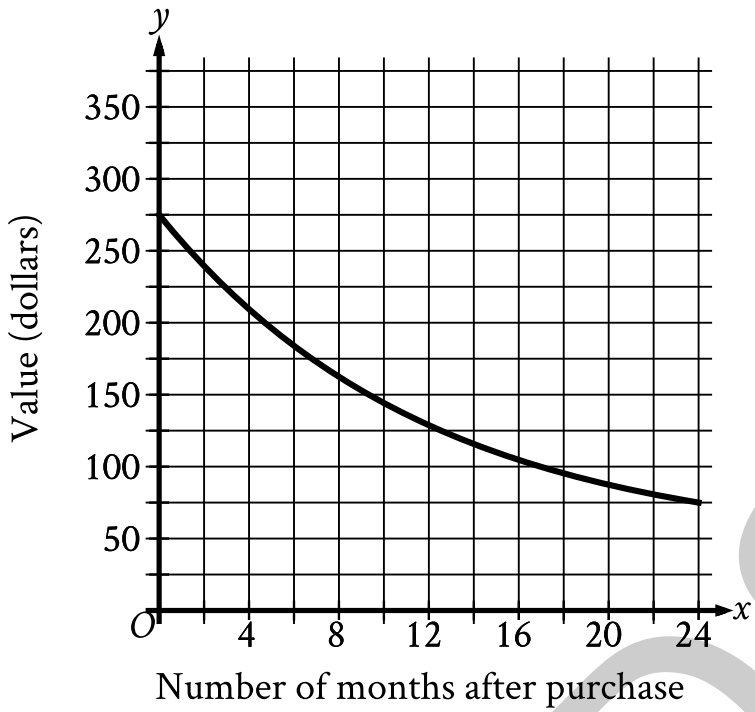
D) $(0, 0)$

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Question 13



- Moving from left to right:
  - The curve is in quadrant 1.
  - The curve trends down gradually.
- The curve passes through the following approximate points:
  - (0 comma 275)
  - (24 comma 75)

The graph shown gives the estimated value, in dollars, of a mountain bike as a function of the number of months since it was purchased. What is the best interpretation of the y-intercept of the graph in this context?

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A) The estimated value of the bike decreased by approximately 2.75% each year after it was purchased.

B) The estimated value of the bike had decreased by $\$275$ in the 24 months after it was purchased.

C) The estimated value of the bike 24 months after it was purchased was $\$275$.

D) The estimated value of the bike was $\$275$ when it was purchased.

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Question 14

x	y
0	70
1	80
2	90

For a linear relationship between x and y, the table gives three values of x and their corresponding values of y. Which equation represents this relationship?

A) $y = x + 10$

B) $y = 10x + 70$

C) $y = 10x$

D) $y = x + 80$

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Question 15

Which expression is equivalent to $72t^2s^3 - 48t^3s$?

A) $4t^2s(18ts - 12s)$

B) $4t^2s(18s^2 - 12t)$

C) $4ts(18s^2 - 12ts)$

D) $4ts(18t^2s^2 - 12t)$

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Question 16

The function f is defined by $f(x) = x^3 + 7$. What is the value of $f(2)$?

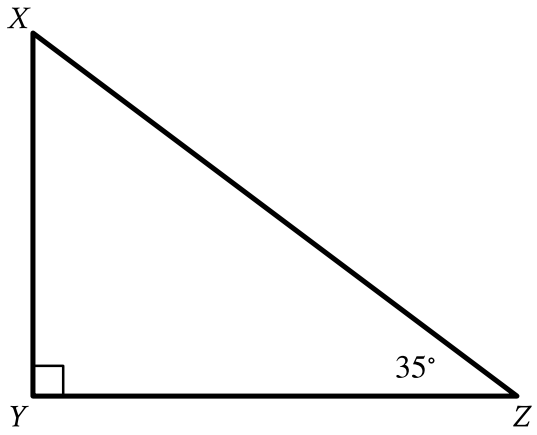
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Question 17



Note: Figure not drawn to scale.

- Angle upper Y is a right angle.
- The measure of angle upper Z is 35°.
- A note indicates the figure is not drawn to scale.

In triangle XYZ, the measure of angle Y is 90 degrees. Triangle RST (not shown) is similar to triangle XYZ such that R, S, and T correspond to X, Y, and Z, respectively. Each side of triangle RST is 19 times the length of the corresponding side of triangle XYZ. What is the measure, in degrees, of one of the angles in triangle RST?

Enter response...

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Question 18

A snack company advertises that bags of pretzels produced by the company weigh, on average, 1 pound each. To test this, Sam selected at random 40 bags of pretzels produced by the company and weighed each bag. Based on his measurements, Sam estimated that for all bags of pretzels produced by the company, the average weight per bag is 1.05 pounds, with a margin of error of 0.06 pounds. For all bags of pretzels produced by the company, which of the following is the most appropriate conclusion about the average weight w, in pounds, per bag?

A) $0.99 \leq w \leq 1.05$

B) $w = 1.05$

C) $1.05 \leq w \leq 1.11$

D) $0.99 \leq w \leq 1.11$

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Question 19

In the *xy*-plane, line k passes through the points $(0, -7)$ and $(1, -1)$. Which equation defines line k?

A) $y = -x + 6$

B) $y = 6x - 7$

C) $y = \frac{1}{6}x - 7$

D) $y = -x + \frac{1}{6}$

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Question 20

The function f is defined by $f(x) = b^x + c$, where b and c are positive constants. In the *xy*-plane, the graph of $y = f(x)$ passes through the point $(0, 25)$. What is the value of $f(0)$?

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Question 21

The measure of angle Z is 45°. What is the measure, in radians, of angle Z?

A) $\frac{1}{2}\pi$

B) $\frac{3}{4}\pi$

C) $\frac{1}{8}\pi$

D) $\frac{1}{4}\pi$

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Question 22

For a certain type of rope, the equation $y = 900ax^2$, where a is a constant, gives the estimated breaking strength y, in pounds, of a rope with a circumference of x inches. Based on this equation, if a rope of this type has a circumference of 1.75 inches, it has an estimated breaking strength of $3{,}858.75$ pounds. What is the estimated breaking strength, in pounds, of a rope of this type that has a circumference of 3.50 inches?

Enter response...

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